

Regression Assignment Answers

1. What is Simple Linear Regression?

Simple Linear Regression (SLR) is a statistical method used to find the relationship between one independent variable (X) and one dependent variable (Y).

The equation is:

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Where:

- Y = Dependent variable
- X = Independent variable
- m = Slope
- c = Intercept

2. What are the key assumptions of Simple Linear Regression?

1. Linear relationship between X and Y
2. Independence of observations
3. Homoscedasticity (constant variance)
4. Normally distributed residuals
5. No significant outliers

3. What does coefficient m represent in Y = mX + c?

In:

`genui{"math_block_widget_always_prefetch_v2":{"content":"Y = mX + c"}}`

The coefficient **m** represents the slope of the line.
It shows how much Y changes when X increases by 1 unit.

4. What does intercept c represent in Y = mX + c?

The intercept **c** represents the value of Y when X = 0.

5. How do we calculate slope m in Simple Linear Regression?

The slope is calculated using:

$$m = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2}$$

6. What is the purpose of the least squares method?

The least squares method minimizes the sum of squared errors between actual and predicted values.

It helps find the best-fit regression line.

7. How is R^2 interpreted in Simple Linear Regression?

R^2 (coefficient of determination) measures how well the model explains variability in data.

- $R^2 = 1 \rightarrow$ Perfect fit
- $R^2 = 0 \rightarrow$ No explanatory power

Example:

$R^2 = 0.85$ means 85% variation is explained by the model.

8. What is Multiple Linear Regression?

Multiple Linear Regression (MLR) predicts a dependent variable using multiple independent variables.

General equation:

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

9. Difference between Simple and Multiple Linear Regression

Simple Linear Regression	Multiple Linear Regression
One independent variable	Multiple independent variables
Simpler model	More complex model
Equation has one X	Equation has many X variables

10. Key assumptions of Multiple Linear Regression

1. Linear relationship
2. Independence of errors
3. Homoscedasticity
4. No multicollinearity
5. Normal distribution of residuals

11. What is heteroscedasticity?

Heteroscedasticity occurs when error variance is not constant across observations.

Effects:

- Unreliable coefficient estimates
- Incorrect statistical tests

12. How can you improve a model with high multicollinearity?

Methods:

1. Remove correlated variables
2. Use feature selection
3. Apply PCA
4. Use Ridge/Lasso Regression

13. Techniques for transforming categorical variables

1. Label Encoding
2. One-Hot Encoding
3. Ordinal Encoding

14. Role of interaction terms in Multiple Linear Regression

Interaction terms show how one variable affects another variable's impact on the target.

Example:

Study Hours $\times$ Sleep Hours

15. Difference in interpretation of intercept in SLR and MLR

- In SLR, intercept is Y when one X = 0.
- In MLR, intercept is Y when all X variables = 0.

16. Significance of slope in regression analysis

Slope indicates:

- Direction of relationship
- Strength of effect

Positive slope $\rightarrow$ direct relationship

Negative slope $\rightarrow$ inverse relationship

17. How does intercept provide context?

The intercept provides a baseline prediction when predictor variables are zero.

18. Limitations of using R^2 alone

1. Does not detect overfitting
2. Cannot measure prediction accuracy fully
3. Always increases with more variables

Adjusted R^2 is often preferred.

19. Interpretation of a large standard error

A large standard error means:

- Coefficient estimates are less reliable
- Predictions may be unstable

20. Identifying heteroscedasticity in residual plots

If residuals show funnel-shaped patterns, heteroscedasticity may exist.

It is important because it violates regression assumptions.

21. High R^2 but low Adjusted R^2 meaning

It means unnecessary variables are added to the model.

The model may be overfitting.

22. Why scale variables in Multiple Linear Regression?

Scaling:

- Improves model performance
- Helps gradient-based optimization
- Prevents dominance of large-scale features

23. What is Polynomial Regression?

Polynomial Regression models nonlinear relationships using polynomial terms.

Example:

$$Y = a + bX + cX^2 + dX^3$$

24. Difference between Polynomial and Linear Regression

Linear Regression	Polynomial Regression
Straight-line relationship	Curved relationship
Simpler	More flexible

Linear equation	Polynomial equation
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25. When is Polynomial Regression used?

It is used when data shows a nonlinear relationship.

Example:

Population growth, stock trends

26. General equation for Polynomial Regression

$$Y = a + b_1X + b_2X^2 + b_3X^3 + \dots + b_nX^n$$

27. Can Polynomial Regression be applied to multiple variables?

Yes, polynomial regression can include multiple variables and polynomial terms.

28. Limitations of Polynomial Regression

1. Overfitting risk
2. Sensitive to outliers
3. Complex interpretation
4. High computational cost for large degree

29. Methods to evaluate polynomial degree

1. R^2 Score
2. Adjusted R^2
3. RMSE
4. Cross-validation
5. Residual analysis

30. Why is visualization important in Polynomial Regression?

Visualization helps:

- Understand nonlinear patterns
- Detect overfitting
- Compare model fit

31. How is Polynomial Regression implemented in Python?

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import make_pipeline
```

```
model = make_pipeline(
```

```
    PolynomialFeatures(degree=2),  
    LinearRegression()  
)
```

```
model.fit(X, y)  
predictions = model.predict(X)
```